

How Much of Turkish Wordle Is Luck?

A Multi-Baseline Skill/Luck Split Using Letter Frequency and Information Theory

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Abstract—This paper measures how much of winning at Turkish Wordle (*wordleturkce.bundle.app*) is due to luck and how much is due to skill. We first extract the full word lists from the live game: 2,936 unique answer words and 5,500 valid guesses. We then simulate four player models on every puzzle in the pool. A player that only follows the colored feedback (green, yellow, gray), but uses no word-selection strategy, already wins 96.97% of the games. This shows that feedback alone carries a large amount of information. However, this player is allowed to know every word in the dictionary, which is not realistic for most humans. We therefore add a vocabulary-limited human model, where the player only knows the top K common words, and an entropy-greedy bot in the style of the *3Blue1Brown* lesson. The entropy bot wins every puzzle with 3.59 guesses on average. With $K=1000$ (covering around 48% of daily puzzles), the measured split is about 34% luck and 66% skill. Inside that skill share, vocabulary explains about 95% and strategy explains about 5%. We also report four other baselines, where the headline luck share changes from 0.1% to 97%. The main message is that a skill/luck number for Wordle is only meaningful after the baseline player is made explicit.

Index Terms—Wordle, information theory, Shannon entropy, Turkish language, word games, skill versus luck, simulation, computational linguistics

I. INTRODUCTION

Wordle was made by Josh Wardle in 2021 and was later bought by The New York Times. In the game, the player has six tries to find a hidden five-letter word. After each guess, the game returns one color for each letter position: green means the right letter is in the right place, yellow means the right letter is in the wrong place, and gray means the letter is not in the word. Repeated letters need one extra counting rule, because the same letter can appear more than once in a guess or answer.

The game became popular because the rules are simple, but the decisions are not. A player must choose between common words, words with many different letters, and words that separate the remaining candidates well. This makes Wordle a useful small test case for information theory, language structure, and human word knowledge. Many

local versions also appeared, including the Turkish-language *wordleturkce.bundle.app*, which uses its own list of Turkish five-letter words.

A common question among players is: how much of a win is luck, and how much is skill? The question sounds simple, but it is not well defined. “Luck” could mean that the daily word was easy. It could also mean that the player happened to know the word, or that the feedback happened to remove many wrong answers. In the same way, “skill” could mean knowing many words, choosing a good opener, or using feedback correctly. A percentage such as “ $X\%$ luck and $Y\%$ skill” therefore needs a baseline: a clear description of what the no-skill or lower-skill player can do.

Most Wordle papers and solver discussions focus on finding better guesses. That is important, but it answers a different question. Solver papers compare algorithms. This paper asks how the answer changes when we compare a strong solver to several weaker players, including a human-like player whose vocabulary is limited. This is the main part of the work that is different from previous Wordle studies: we do not only ask which solver is best; we also measure how much of the gap can be explained by vocabulary, feedback, and strategic word choice.

The Turkish version is also worth studying on its own. Turkish word forms and letter positions are not the same as English. Five-letter Turkish words often place vowels and consonants in different positions than English words, and common Turkish words do not have the same letter frequency pattern. For that reason, results from English Wordle should not be copied directly to Turkish Wordle without checking the data.

This paper answers the question with five steps:

- 1) Extract the answer pool and guess dictionary from the shipped JavaScript bundle (Section III);
- 2) Build four player models, from random guessing to an information-theoretic bot (Sections IV, V);
- 3) Add a human-like player whose vocabulary is limited to the top K common words (Section IV-D);
- 4) Compute the skill/luck split under five baselines, so the baseline sensitivity is visible (Section VII); and

- 5) Study per-word difficulty and the word families that remain hard even for the entropy bot (Section VIII).

II. RELATED WORK

Wordle solvers got a lot of attention soon after the game came out. Sanderson’s video [2] made the entropy-greedy approach widely known. It uses Shannon’s information theory [1]: each turn the player picks the guess with the highest expected information gain, $H(g \mid C) = -\sum_p P(p \mid g, C) \log_2 P(p \mid g, C)$. Later work showed that the truly optimal English bot needs about 3.42 guesses on average and wins every game by doing full game-tree search [3]. A recent peer-reviewed paper by Aladaileh et al. [8] confirms again that entropy beats simple letter-distribution heuristics on English Wordle.

For Turkish Wordle, Akbulut and Senturk [13] already made the first careful solver study. They compare LSTM, MaxEnt, and rule-based methods on 1,000 Turkish target words. They also show that linguistic features, especially dense CVCVC templates and low word frequency, help predict word difficulty. This paper uses that result as background, but it asks a different research question. We do not try to beat other solvers. Instead, we ask how much of a win remains after we choose a baseline player. We also use the full 2,936-word solution pool from the deployed game and add a human-like player with limited vocabulary.

The vocabulary-limited idea has a close relative in the HSW model of Qiu and Zhong [14], who use a vocabulary-population (VP) component based on a normal-distribution assumption to simulate heterogeneous English players. Our model is different: we do not assume any distribution. Instead we rank words by how often the deployed bundle picks them as the daily answer.

For games in general, the skill-vs-luck question has been studied more formally. The most rigorous work is on poker [5], which uses a predictive-likelihood argument. We follow a simpler counterfactual idea from Mauboussin [4]: a game outcome counts as skill if the skilled agent wins where the baseline agent would lose, and as luck if both win or both lose. For English Wordle, Dilger [15] applies information theory to a large sample of player data and uses the share of one-guess solves to estimate a cheating rate. Our work is different: we report the skill/luck split under several human baselines, not a single score against one bot.

This leaves a gap that the present paper targets. We are not aware of earlier Wordle work that combines three pieces in one experiment: a multi-baseline skill/luck split, a vocabulary-limited human baseline, and the full deployed Turkish Wordle solution pool. This matters because Turkish is not just English with different letters. It is agglutinative, and many five-letter Turkish words follow CVCVC-like shapes. Therefore the letter-position counts, good openers, and hard word families must be checked on Turkish data directly.

III. DATA AND METHODOLOGY

A. Word Lists

The game runs fully in the browser. The complete word list ships inside one JavaScript file at <https://wordleturkce.bundle.app/static/main.2023-04-23.js>. We find two arrays in it:

- *La*: 11,470 entries, but only 2,936 unique. The daily answer is $La[i \bmod 11470]$, where i is the day index. The same word can appear many times—for example *buruk* and *selam* each appear 8 times, while rare words appear once. We read this repetition as a commonality signal: the curator made common words more likely to be the daily answer.
- *Ta*: 5,531 entries, 5,499 unique. These count as valid guesses but never appear as the answer.

After we drop duplicates, the full guess dictionary has 5,500 unique words. The daily answer comes from the 2,936 unique solutions, with the *La* multiplicities used as weights.

B. Feedback Function

We implement the standard Wordle feedback rule, including the case with repeated letters. It is a deterministic function $\text{feedback} : \Sigma^5 \times \Sigma^5 \rightarrow \{0, 1, 2\}^5$, where 0, 1, and 2 stand for gray, yellow, and green. We assign greens first. Then we walk through the non-green positions of the guess from left to right and mark a yellow when the answer still has an unmatched copy of that letter. We encode each five-cell pattern as a base-3 integer in $[0, 242]$ so that we can index it directly.

C. Letter Frequency Analysis

Over the weighted 11,470-entry solution distribution, the five most common letters are A (13.04%), E (8.52%), İ (6.90%), K (6.81%), and R (6.09%). Together they cover 41% of all letter occurrences. By position:

TABLE I
MOST FREQUENT LETTERS BY POSITION (TOP 5 EACH).

Position	Top 5 letters (with counts)
1	K (1336), S (1121), T (841), B (751), M (714)
2	A (3199), E (1881), İ (1142), O (938), U (782)
3	R (1435), L (1183), K (799), N (757), M (747)
4	A (2308), İ (1587), E (1530), I (868), U (843)
5	K (1367), N (1189), R (1070), A (1052), E (971)

This pattern matches Turkish phonotactics. Turkish words use a canonical $(C)V(C)$ syllable template [12], and CVCVC is one of the most common five-letter shapes. Vowels are mostly in positions 2 and 4 and consonants in positions 1, 3, and 5.

IV. PLAYER MODELS

We use four player models. All of them follow the standard rules: up to six guesses, and each guess must be a valid five-letter word from the dictionary.

A. Pure Random ($\mathcal{P}_{\text{rand}}$)

At each turn this player picks a random word from the 5,500-word dictionary and ignores all feedback. This is the zero-information baseline. The expected win rate in six guesses is $1 - (1 - 1/5500)^6 \approx 0.109\%$.

B. Consistent Random ($\mathcal{P}_{\text{cons}}$)

This player keeps the set of candidates that are still consistent with the feedback so far. At each turn it picks a random word from that set. It follows the rules perfectly but uses no strategy beyond that.

C. Frequency Bot ($\mathcal{P}_{\text{freq}}$)

At each turn this bot recomputes positional letter frequencies on the current candidate set. Each candidate w receives a score $\text{score}(w) = \sum_i \mathbb{I}[w[i] \notin w[0:i]] \cdot \text{posfreq}[i, w[i]]$, which sums the contributions of the *distinct* letters only. The bot picks the candidate with the highest score.

D. Vocabulary-Limited Human ($\mathcal{P}_{\text{hum},K}$)

This model represents a casual human player by a vocabulary size K . We rank words by how often La repeats them (the commonality proxy), and break ties alphabetically. The top K are the player’s vocabulary. At each turn, the player picks a random word from the intersection of the remaining candidates and the vocabulary. If that intersection is empty, the player picks a random word from the vocabulary as a forced “probe” guess.

V. ENTROPY-GREEDY BOT

To get a tight skill ceiling we add an entropy-greedy bot $\mathcal{P}_{\text{ent}}$ in the style of [2]. For a guess g and candidate set C , the partition that g creates on C is $\{C_p : p \in \{0, \dots, 242\}\}$ where $C_p = \{a \in C : \text{feedback}(g, a) = p\}$. The expected information gain is the entropy of this partition:

$$H(g | C) = - \sum_p \frac{|C_p|}{|C|} \log_2 \frac{|C_p|}{|C|}. \quad (1)$$

The bot picks $g^* = \arg \max_g H(g | C)$ and, when several guesses are tied, prefers a guess that is itself in C .

A. Implementation

A direct implementation needs $O(|G| \cdot |C|)$ feedback evaluations per turn, which is too slow when we simulate thousands of puzzles. To speed this up we precompute a pattern matrix $M \in \{0, \dots, 242\}^{|G| \times |A|}$ once. We store the encoded feedback for every guess-answer pair as a `uint8`. The matrix takes about 8MB and is built in about 0.5s using vectorized NumPy [9] operations that work in parallel over letters.

To check the fast feedback function we compared it with a plain Python reference on 5,000 random word pairs. We found 0 mismatches.

TABLE II
PERFORMANCE OF ALL FOUR PLAYER MODELS ON THE 2,936-WORD SOLUTION POOL (UNIFORM SAMPLING).

Player	Win rate	Avg. guesses	Opener
$\mathcal{P}_{\text{rand}}$ (pure random)	0.10%	4.33	—
$\mathcal{P}_{\text{hum},1000}$ ($K=1000$)	33.92%	3.84	—
$\mathcal{P}_{\text{cons}}$ (consistent)	96.97%	4.21	—
$\mathcal{P}_{\text{freq}}$ (frequency)	99.18%	3.77	salik
$\mathcal{P}_{\text{ent}}$ (entropy)	100.00%	3.59	merak

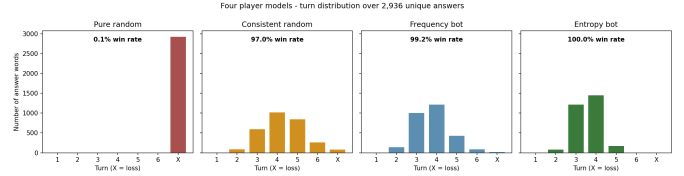


Fig. 1. Turn counts for the four player models on 2,936 puzzles. Pure random almost always runs out of guesses. Consistent random and the frequency bot have long tails. The entropy bot is tightly grouped and never goes past five guesses.

B. Results

The best entropy opener is merak with $H = 5.81$ bits. The top entropy openers include merak, کنار, kamer, kelam, kalem, malik, karne, katre, keman, and karni. This list shows that mid-frequency liquid consonants (R, L, N, M) can be more useful than the very frequent letter A, which dominated the frequency-score top picks.

On all 2,936 puzzles, the entropy bot wins 100.00% of the time, with 3.59 guesses on average. It never reaches turn six. The turn histogram is $\{1 : 1, 2 : 84, 3 : 1221, 4 : 1455, 5 : 175\}$.

VI. SIMULATION RESULTS

Table II shows the main numbers. Fig. 1 shows the turn distribution for each model. The largest gap is between $\mathcal{P}_{\text{rand}}$ (0.1%) and $\mathcal{P}_{\text{cons}}$ (96.97%). Once a player uses the feedback correctly, most of the win rate is already recovered. Strategy on top of that adds a smaller gain.

A. Vocabulary Sensitivity

Table III and Fig. 2 show how the win rate depends on the vocabulary size K . The empirical fit

$$P(\text{win} | K) \approx \text{coverage}(K) \cdot 0.97 \quad (2)$$

holds for every K we tested, within sampling noise. Equation 2 gives a direct interpretation: *knowing the word is almost necessary and sufficient to win*. If the day’s word is in the player’s vocabulary, consistent-random play wins with probability about 0.97. Otherwise the player usually loses.

VII. SKILL/LUCK DECOMPOSITION

We use a counterfactual decomposition. The entropy bot’s wins count as *luck* when the baseline player would have won too, and as *skill* when the baseline player would have lost. For a baseline $\mathcal{P}_b$ with win rate w_b and a ceiling $\mathcal{P}_{\text{ent}}$ with win

TABLE III

WIN RATES OF THE VOCABULARY-LIMITED HUMAN MODEL AND THE SHARE OF DAILY PUZZLES ITS VOCABULARY COVERS, AS A FUNCTION OF K .

K	Uniform win %	Weighted win %	Coverage %
200	6.81	10.92	10.92
500	17.00	25.74	25.79
1000	33.92	47.39	47.59
1500	50.72	67.00	67.50
2000	67.13	83.68	84.93
2500	83.58	94.46	96.20
2936	97.21	97.20	100.00

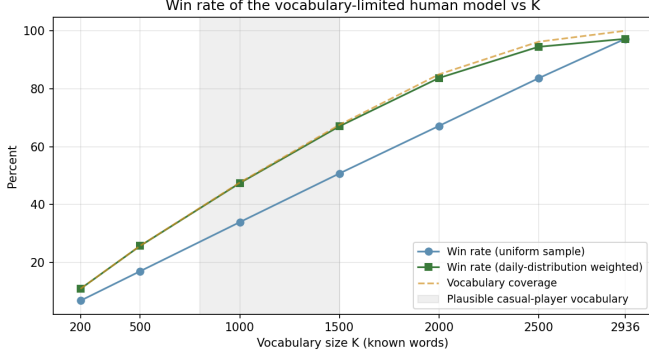


Fig. 2. Win rate of the vocabulary-limited human model as a function of vocabulary size K . The weighted curve is almost identical to the vocabulary coverage curve, which matches Eq. 2. The shaded band is the plausible range for typical casual players.

rate $w_c = 1$, the luck share is w_b/w_c and the skill share is $(w_c - w_b)/w_c$.

Table IV and Fig. 3 show the split. The headline answer depends entirely on the baseline we pick. We argue that the most realistic baseline for an average human is $\mathcal{P}_{\text{hum}, K \approx 1000}$, since informal estimates put active Turkish five-letter vocabulary around this size for casual players. Under this baseline, **$\approx 66\%$ of Turkish Wordle wins come from skill and $\approx 34\%$ come from luck.**

A. Skill is Mostly Vocabulary

Equation 2 also tells us that the skill share itself splits into two parts. Going from $\mathcal{P}_{\text{hum}, 1000}$ to $\mathcal{P}_{\text{cons}}$ (perfect vocabulary, no strategy) adds 63 points of win rate (33.92% to 96.97%). Going from $\mathcal{P}_{\text{cons}}$ to $\mathcal{P}_{\text{ent}}$ (strategy on top of perfect vocabulary) adds only 3 points more (96.97% to 100.00%). So out of the 66-point skill gap, vocabulary explains about 95% and strategic word selection explains about 5%.

This differs from the usual story in popular Wordle posts, which focuses on opener choice and information-theoretic optimization. Our result suggests that *lexical knowledge* matters much more than strategic sophistication for who wins more often.

VIII. PER-WORD DIFFICULTY

Fig. 4 shows how many turns the entropy bot needs to solve each of the 2,936 puzzles. The most common turn count is 4.

TABLE IV

SKILL/LUCK SPLIT UNDER FIVE DIFFERENT BASELINES. THE RESULT CHANGES BY ALMOST TWO ORDERS OF MAGNITUDE.

Baseline $\mathcal{P}_b$	w_b	Luck %	Skill %
Pure random	0.10%	0.1%	99.9%
Human ($K=500$)	17.00%	17.0%	83.0%
Human ($K=1000$)	33.92%	33.9%	66.1%
Human ($K=2000$)	67.13%	67.1%	32.9%
Consistent random	96.97%	97.0%	3.0%

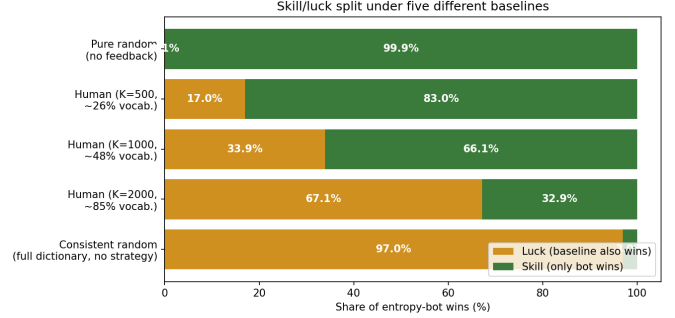


Fig. 3. Skill and luck shares of entropy-bot wins under five baselines. The answer to “how much of Wordle is luck” moves from 0.1% to 97% just based on which baseline we pick.

The bot never reaches turn 6. The puzzles that take 5 turns (175 words) are the structurally hardest ones.

When we look inside the 5-turn group we see a common pattern: large families of look-alike words that differ in only one position, for example KAPAK / KAÇAK / KAVAK / KAZAK. Once the candidate set shrinks to such a family, even an optimal bot has to test the remaining words one by one because each guess tells it about one position only. This is an irreducible source of variance. No strategy can remove it, and an ideal human would face the same problem.

This kind of luck is *structural*, and is different from the player-level luck we measured earlier. Even a perfect-information player has to spend more turns when the day’s word lives in a dense look-alike family.

IX. DISCUSSION

A. Definitional Sensitivity

The main methodological point of this work is that the popular “ $X\%$ luck, $Y\%$ skill” framing is not well defined without a baseline. The same data, looked at under five reasonable baselines, gives skill shares anywhere between 3% and 99.9%. We therefore think single-number claims of this kind should be avoided in both popular and academic discussion of game design.

B. Vocabulary as the Dominant Skill

The near-identity between vocabulary coverage and win rate (Eq. 2) points to an under-discussed finding: Wordle is mostly a vocabulary test. Strategy helps, but its marginal effect is small. A player who wants a higher win rate may gain more by learning more words than by changing to a more complex

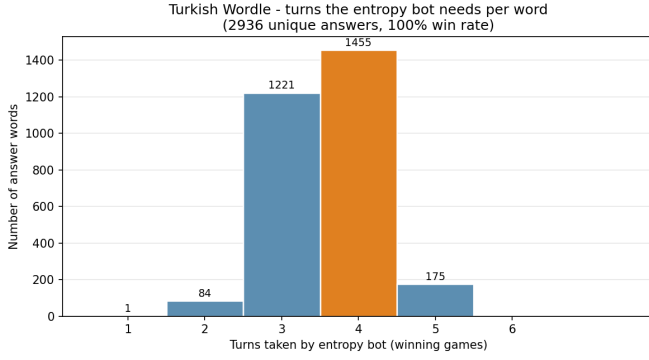


Fig. 4. Turn counts of the entropy-greedy bot on all 2,936 solutions. No puzzle needs 6 turns; the hardest 175 take 5.

opening word. This matches independent observational work by Liang et al. [11], who show that real Wordle players narrow candidates using semantic, orthographic, and morphological similarities, not information-theoretic optimality.

C. Information-Theoretic Bound

The Shannon entropy of the weighted daily distribution is 11.36 bits. The theoretical ceiling per guess is $\log_2 243 = 7.92$ bits. The entropy bot pulls out about 3.16 bits per guess on average (the same as needing 3.59 guesses to remove 11.36 bits of uncertainty). That is around 40% of the per-guess ceiling. The gap has two sources: (i) the greedy choice is not the same as full lookahead, and (ii) every guess must be a real word, not just any five-character string.

D. Limitations

The vocabulary model uses bundle multiplicity as a proxy for commonality. A more rigorous human model would calibrate against an external Turkish corpus and would also include imperfect feedback tracking, especially for words with repeated letters. Stronger solver models, such as the Bertsimas–Paskov-style full-tree search [3], would raise the skill ceiling slightly but would not change the overall split.

X. CONCLUSION

We have presented a multi-baseline analysis of skill versus luck in Turkish Wordle, built on a full extraction of the deployed game’s solution and dictionary lists. Four player models were simulated on all 2,936 puzzles: the entropy-greedy ceiling wins every game, while a realistic vocabulary-limited human ($K=1000$) wins about 34%. Under this baseline, the skill share of wins is about 66%, and about 95% of that is vocabulary while only about 5% is strategy. We stress that the headline split is very sensitive to the baseline and we report a five-baseline table to back up this point.

All code, simulation outputs, intermediate data, and reproduction steps are available as open source at the project repository.

REPRODUCIBILITY

The source code, extracted word lists, full simulation outputs, and reproduction steps are available in the project’s open-source repository. The whole pipeline runs end-to-end in under three minutes on a normal laptop, using only NumPy [9] and Matplotlib [10].

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